Metro Ticket Generating System

Phase 4-Data Migration, Testing and Security

Data Handling

Data handling refers to how information collected from users is stored, processed, and managed within the system. In the Metro Ticket Generating System, data handling was implemented to ensure that all ticket booking details entered by users through the Service Catalog are properly captured and stored in the backend database.

The system uses ServiceNow Flow Designer to automatically retrieve catalog variables submitted by the user and store them in a custom database table called Metro Ticket (u_metro_ticket). This ensures that every ticket booking request creates a structured record in the database for tracking and management.

Proper data handling ensures that user data is processed efficiently, stored securely, and can be retrieved whenever required.

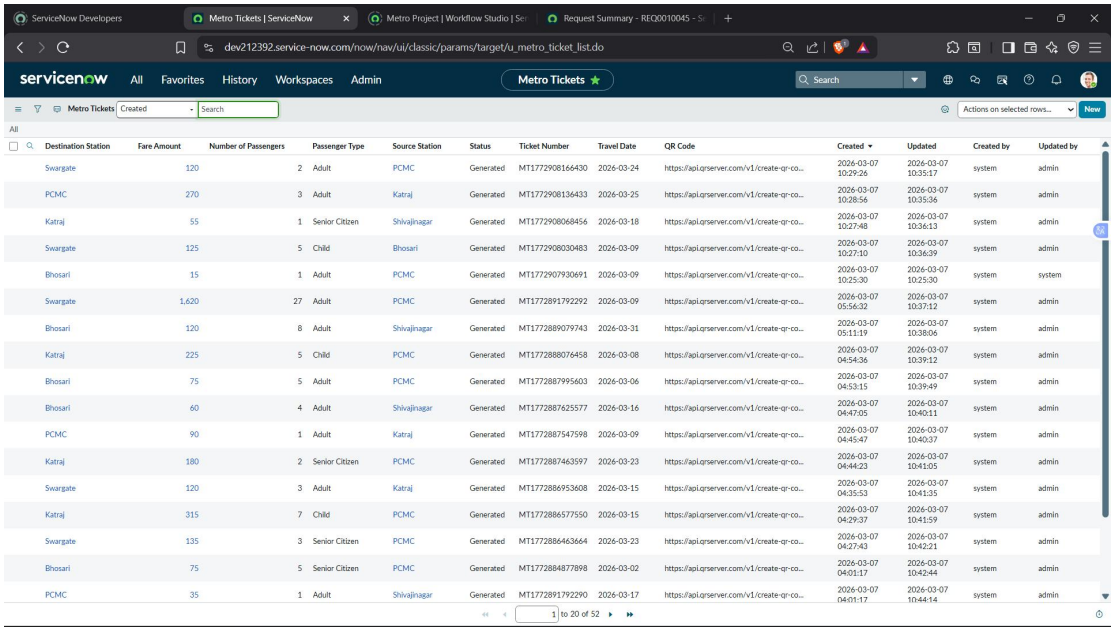
Ticket Data Storage in Custom Database Table

When a user submits a ticket booking request, the data entered in the Service Catalog form is captured and stored in the Metro Ticket database table.

The table stores important ticket information such as:

Ticket Number
Source Station
Destination Station
Passenger Type
Number of Passengers
Travel Date
Fare Amount
Ticket Status
QR Code

Each ticket booking creates a new record in this table, allowing administrators to monitor and manage ticket information easily.



Destination Station	Fare Amount	Number of Passengers	Passenger Type	Source Station	Status	Ticket Number	Travel Date	QR Code	Created	Updated	Created by	Updated by
Swargate	120	2	Adult	PCMC	Generated	MT1772908166430	2026-03-24	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 10:29:26	2026-03-07 10:35:17	system	admin
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Swargate	125	5	Child	Bhosari	Generated	MT1772908030483	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 10:27:10	2026-03-07 10:36:39	system	admin
Bhosari	15	1	Adult	PCMC	Generated	MT1772907930691	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 10:25:30	2026-03-07 10:25:30	system	system
Swargate	1,620	27	Adult	PCMC	Generated	MT1772891792292	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 05:56:32	2026-03-07 10:37:12	system	admin
Bhosari	120	8	Adult	Shivajinagar	Generated	MT1772889079743	2026-03-31	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 05:11:19	2026-03-07 10:38:06	system	admin
Katraj	225	5	Child	PCMC	Generated	MT1772888076458	2026-03-08	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:54:36	2026-03-07 10:39:12	system	admin
Bhosari	75	5	Adult	PCMC	Generated	MT1772887995603	2026-03-06	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:53:15	2026-03-07 10:39:49	system	admin
Bhosari	60	4	Adult	Shivajinagar	Generated	MT1772887625577	2026-03-16	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:47:05	2026-03-07 10:40:11	system	admin
PCMC	90	1	Adult	Katraj	Generated	MT1772887547598	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:45:47	2026-03-07 10:40:37	system	admin
Katraj	180	2	Senior Citizen	PCMC	Generated	MT1772887463597	2026-03-23	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:44:23	2026-03-07 10:41:05	system	admin
Swargate	120	3	Adult	Katraj	Generated	MT1772886953608	2026-03-15	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:35:53	2026-03-07 10:41:35	system	admin
Katraj	315	7	Child	PCMC	Generated	MT1772886577550	2026-03-15	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:29:37	2026-03-07 10:41:59	system	admin
Swargate	135	3	Senior Citizen	PCMC	Generated	MT1772886463664	2026-03-23	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:27:43	2026-03-07 10:42:21	system	admin
Bhosari	75	5	Senior Citizen	PCMC	Generated	MT1772884877898	2026-03-02	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:01:17	2026-03-07 10:42:44	system	admin
PCMC	35	1	Adult	Shivajinagar	Generated	MT1772891792290	2026-03-17	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:01:17	2026-03-07 10:44:14	system	admin

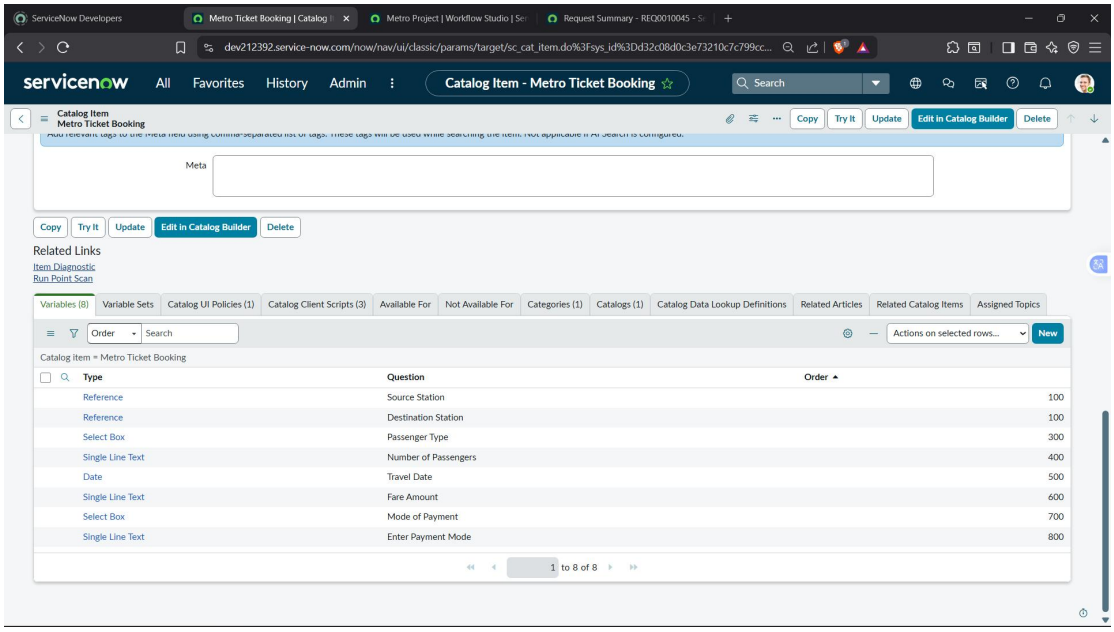
Capturing Catalog Variables

The user enters travel details through the Metro Ticket Booking catalog form. These values are stored as catalog variables.

Examples of captured variables include:

Source Station
Destination Station
Passenger Type
Number of Passengers
Travel Date
Fare Amount
Mode of Payment

These variables are automatically passed to the backend automation when the ticket request is submitted.



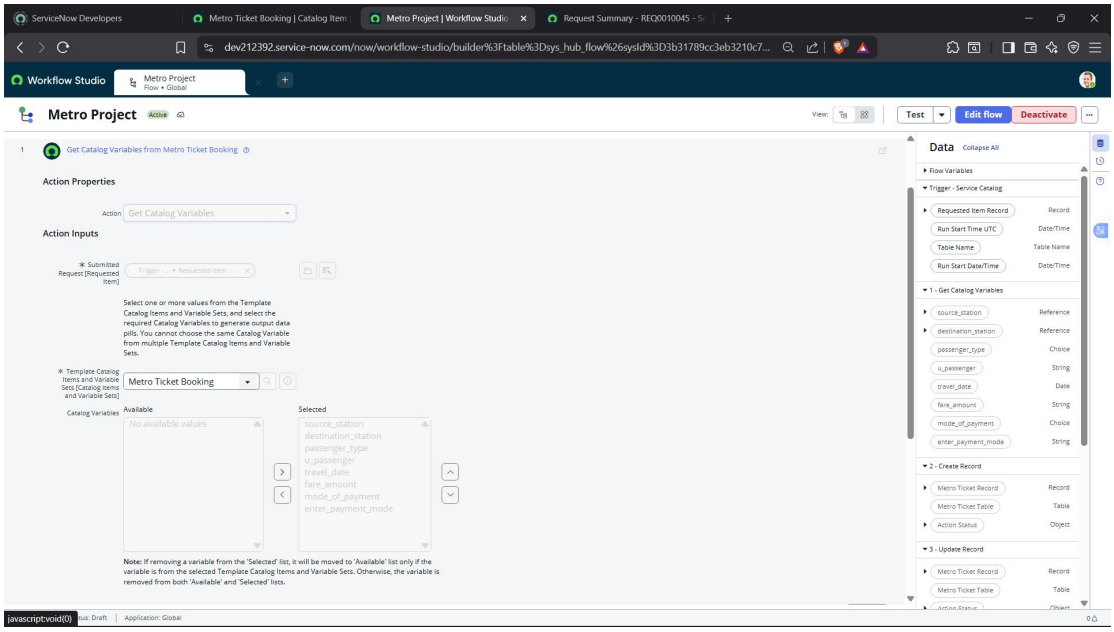
Mapping Variables to Custom Table Records

To store the catalog variables into the Metro Ticket table, Flow Designer automation was used.

The flow performs the following steps:

The flow is triggered when the Metro Ticket Booking catalog item is requested.
The Get Catalog Variables action retrieves all form values submitted by the user.
The Create Record action creates a new record in the Metro Ticket table.
The catalog variables are mapped to the corresponding table fields.
The system stores the ticket information in the database.

This process ensures that all ticket booking information is automatically transferred from the user interface to the backend database.



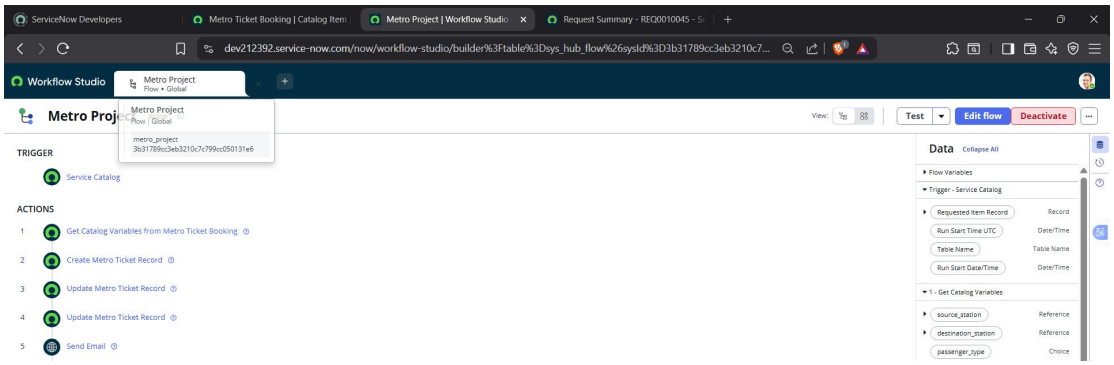
Flow Automation for Data Storage

The automated flow in ServiceNow handles the entire data processing workflow.

The flow consists of the following steps:

Service Catalog Trigger
Retrieve Catalog Variables
Create Metro Ticket Record
Map variables to table fields
Store ticket data in database

This automated process ensures that every ticket booking request results in a properly structured database record without manual intervention.



Benefits of Data Handling Implementation

The data handling mechanism implemented in the Metro Ticket System provides several benefits:

Ensures structured storage of ticket booking data
Reduces manual data entry errors
Automates ticket record creation
Enables easy retrieval and management of ticket records
Supports reporting and ticket tracking

This implementation ensures that the system maintains reliable and organized ticket data throughout the booking process.

Access Control

Access control is an important security mechanism that ensures that only authorized users can access or modify system data. In the Metro Ticket Generating System, role-based access control was implemented to protect the ticket database and restrict unauthorized access.

The system uses ServiceNow Access Control Lists (ACLs) to manage permissions for different users. These rules define who can read, create, update, or delete records in the Metro Ticket database.

Through this implementation, passengers can book tickets using the Service Catalog interface, but they cannot directly access the backend database tables. Only authorized administrators or system users can view or modify database records.

Role-Based Access Implementation

The system follows a role-based access model, where different users have different levels of permissions.

In this project:

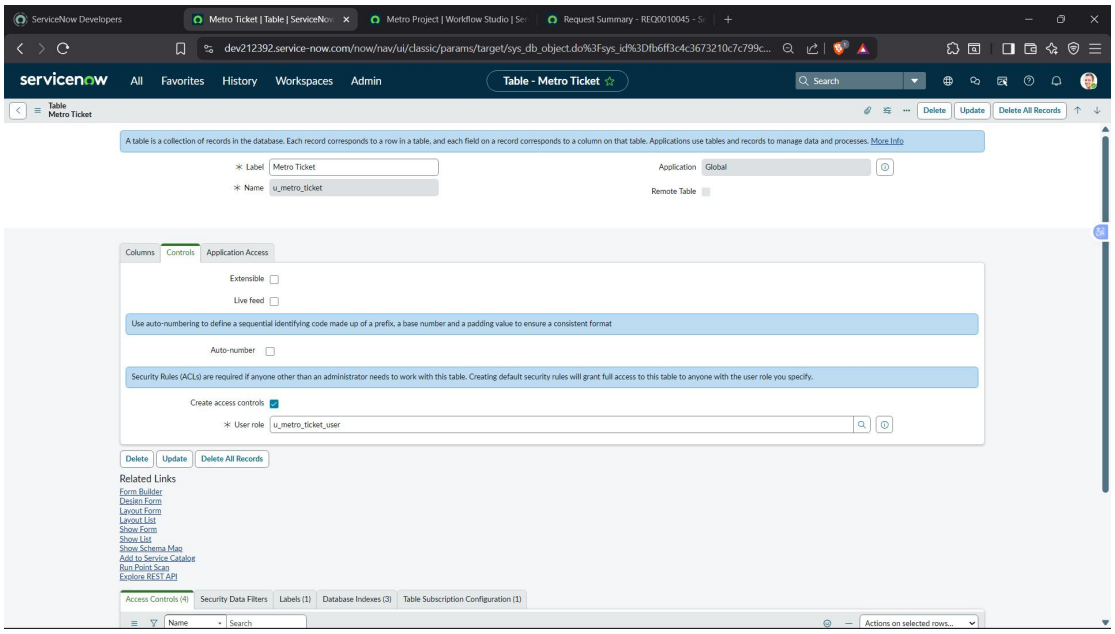
End Users (Passengers)

Can access the Service Portal
Can submit ticket booking requests
Cannot directly view or modify database tables

Administrators / Developers

Can access the Metro Ticket database table
Can manage records and configurations
Can monitor ticket transactions

This ensures that sensitive ticket data is protected while still allowing users to book tickets easily.



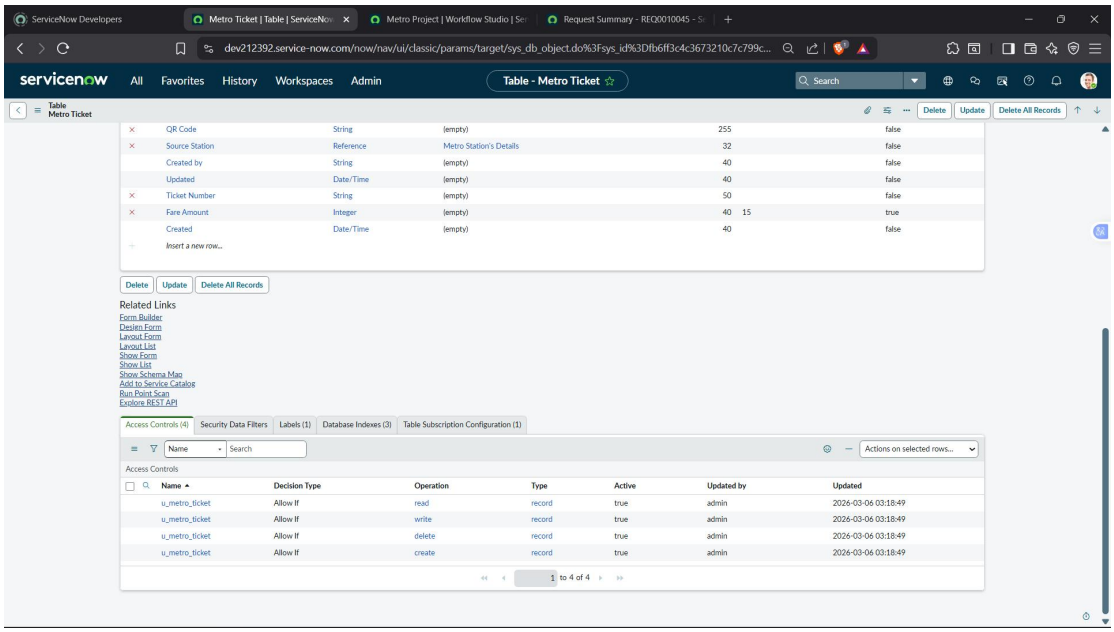
Access Control Lists (ACLs)

When a custom table is created in ServiceNow, the platform automatically generates default Access Control Lists (ACLs) for that table.

In the Metro Ticket database table, the following ACL operations were available:

Read – Allows viewing of records
Create – Allows creation of new records
Write – Allows modification of existing records
Delete – Allows deletion of records

These ACL rules were used to control access permissions and ensure that only authorized roles can interact with the database.



Restricting Direct Database Access

To maintain system security, direct access to the Metro Ticket table was restricted.

Users interact with the system through the Service Catalog interface, which allows them to submit booking requests. The backend Flow Designer automation processes the request and stores the data in the database automatically.

This design prevents users from directly editing database records and ensures controlled data access through predefined workflows.

Benefits of Access Control Implementation

Implementing access control provides several security benefits:

Protects sensitive ticket data from unauthorized users
Prevents accidental modification or deletion of records
Ensures controlled interaction with the database
Maintains system integrity and security
Enables administrators to manage ticket records safely

This structured access control model ensures that the Metro Ticket System operates securely while allowing users to book tickets efficiently.

QA Testing

Quality Assurance (QA) testing is performed to verify that the Metro Ticket Generating System functions correctly and meets the project requirements. The purpose of QA testing is to ensure that all components of the system work properly, including the Service Catalog form, automation workflows, fare calculation logic, ticket generation, and QR code functionality.

During testing, different scenarios were executed to validate that the system behaves as expected when users book a metro ticket. The testing process helped identify and resolve issues related to form validation, automation logic, database updates, and QR code generation.

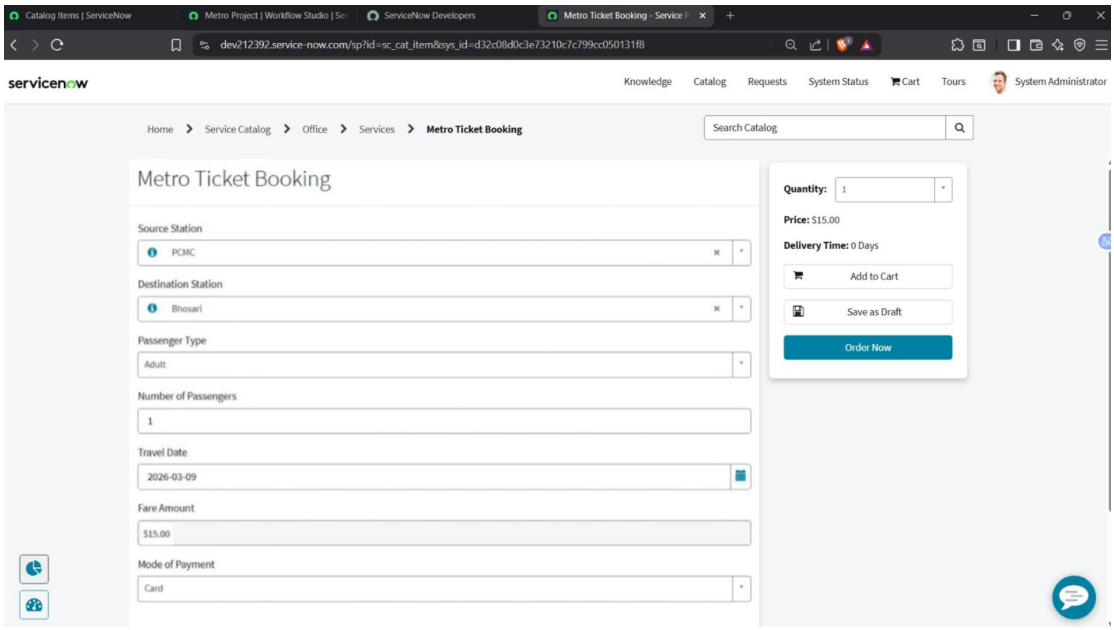
Catalog Form Testing

The first step of testing focused on validating the Service Catalog ticket booking form. The form was tested to ensure that all required fields were visible and functioning correctly.

The following fields were tested:

Source Station
Destination Station
Passenger Type
Number of Passengers
Travel Date
Fare Amount

Each field was verified to ensure that users can correctly input data and that the form behaves properly when values are selected or changed.



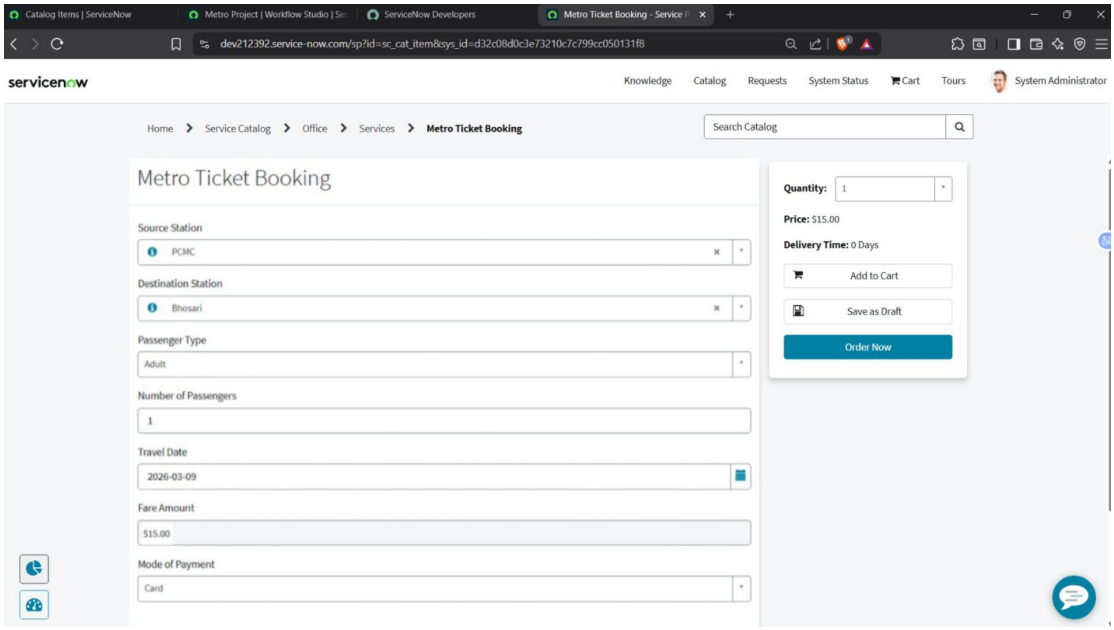
Fare Calculation Validation

The fare calculation logic was tested to ensure that the correct ticket price is generated based on the selected source and destination stations.

Test cases included:

Selecting different station combinations

Changing passenger type (Adult, Child, Senior Citizen)
Adjusting the number of passengers
The system successfully calculated and displayed the correct fare values according to the predefined fare logic.



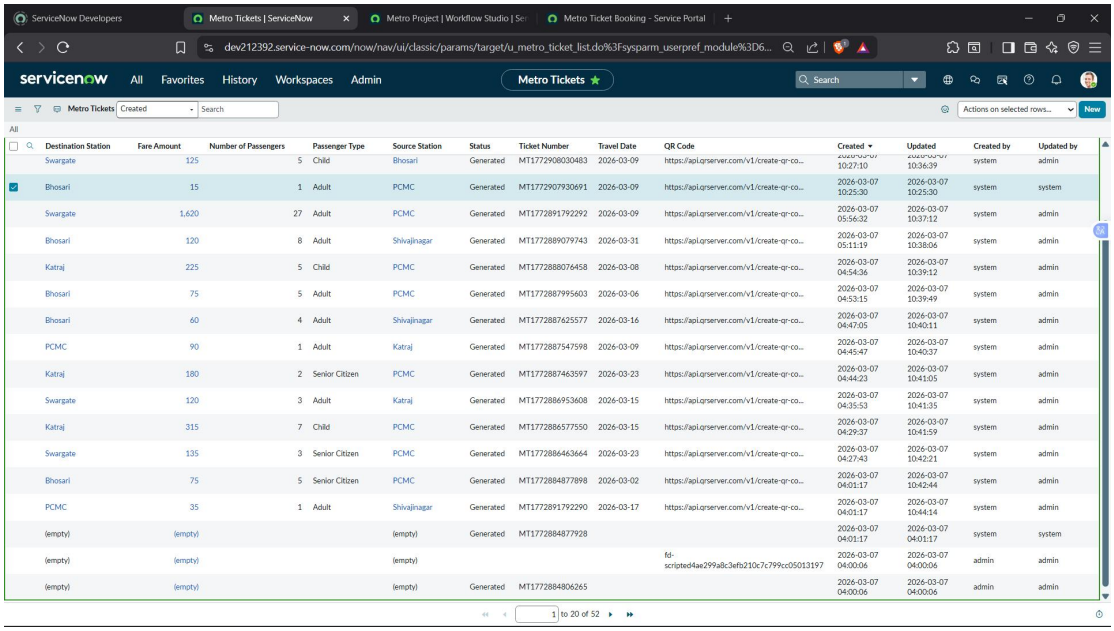
Ticket Generation Testing

After submitting the booking request, the system was tested to confirm that a new ticket record is automatically created in the Metro Ticket database table.

The following were verified:

Ticket Number is generated automatically
Passenger details are correctly stored
Source and destination stations are recorded
Fare amount is saved in the database
Ticket status is updated to Generated

This ensures that every ticket booking request creates a properly stored record in the system.



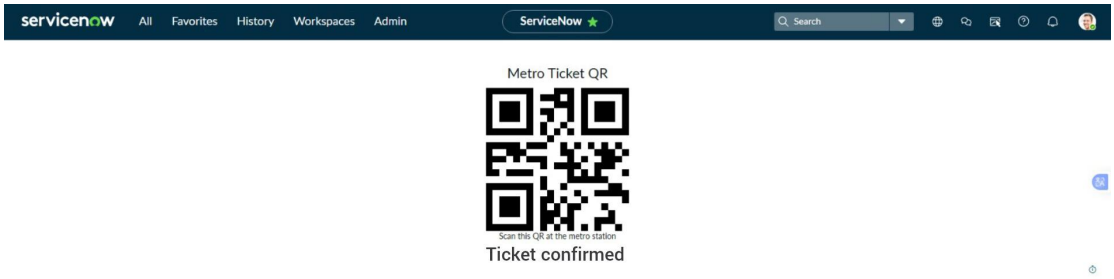
Destination Station	Fare Amount	Number of Passengers	Passenger Type	Source Station	Status	Ticket Number	Travel Date	QR Code	Created	Updated	Created by	Updated by
Swargate	125	5	Child	Bhosari	Generated	MT1772908030483	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	10:27:10	10:36:39	system	admin
Bhosari	15	1	Adult	PCMC	Generated	MT1772907930691	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 10:25:30	2026-03-07 10:25:30	system	system
Swargate	1,620	27	Adult	PCMC	Generated	MT1772891792292	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 05:56:32	2026-03-07 10:37:12	system	admin
Bhosari	120	8	Adult	Shivajinagar	Generated	MT1772889079743	2026-03-31	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 05:11:19	2026-03-07 10:38:06	system	admin
Katraj	225	5	Child	PCMC	Generated	MT1772888076458	2026-03-08	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:54:36	2026-03-07 10:39:12	system	admin
Bhosari	75	5	Adult	PCMC	Generated	MT1772887995603	2026-03-06	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:53:15	2026-03-07 10:39:49	system	admin
Bhosari	60	4	Adult	Shivajinagar	Generated	MT1772887625577	2026-03-16	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:47:05	2026-03-07 10:40:11	system	admin
PCMC	90	1	Adult	Katraj	Generated	MT1772887547598	2026-03-09	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:45:47	2026-03-07 10:40:37	system	admin
Katraj	180	2	Senior Citizen	PCMC	Generated	MT1772887463597	2026-03-23	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:44:23	2026-03-07 10:41:05	system	admin
Swargate	120	3	Adult	Katraj	Generated	MT1772886953608	2026-03-15	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:35:53	2026-03-07 10:41:35	system	admin
Katraj	315	7	Child	PCMC	Generated	MT1772886577550	2026-03-15	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:29:37	2026-03-07 10:41:59	system	admin
Swargate	135	3	Senior Citizen	PCMC	Generated	MT1772886463664	2026-03-23	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:27:43	2026-03-07 10:42:21	system	admin
Bhosari	75	5	Senior Citizen	PCMC	Generated	MT1772884877898	2026-03-02	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:01:17	2026-03-07 10:42:44	system	admin
PCMC	35	1	Adult	Shivajinagar	Generated	MT1772891792290	2026-03-17	https://api.qrserver.com/v1/create-qr-co...	2026-03-07 04:01:17	2026-03-07 10:44:14	system	admin
(empty)	(empty)			(empty)	Generated	MT1772884877928			2026-03-07 04:01:17	2026-03-07 04:01:17	system	system
(empty)	(empty)			(empty)				fd-scripted4ae299a8c3efb210c7c799cc05013197	2026-03-07 04:00:06	2026-03-07 04:00:06	admin	admin
(empty)	(empty)			(empty)	Generated	MT1772884806265			2026-03-07 04:00:06	2026-03-07 04:00:06	admin	admin

QR Code Generation Testing

The QR code generation feature was tested to ensure that a valid QR code is generated after the ticket booking process.

Once the user submits the ticket request:

The system generates a ticket number.
A QR code is generated using the ticket information.
The QR code is displayed in a popup window for the user.
This QR code can be scanned at the metro station to verify the ticket.



End-to-End Workflow Testing

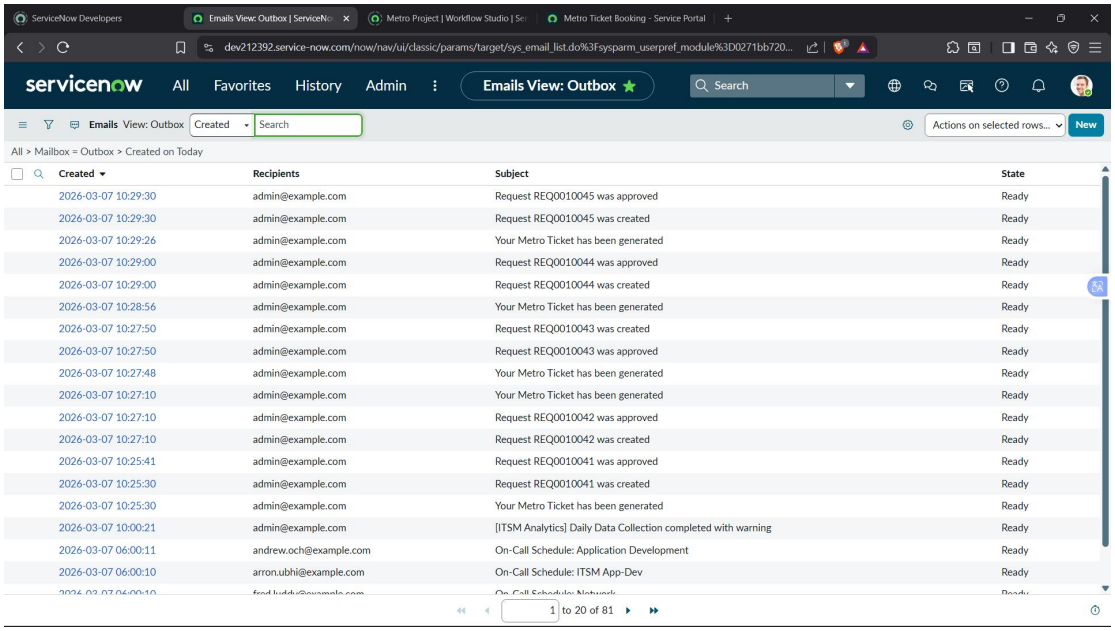
Finally, the complete workflow of the system was tested from start to finish to ensure smooth operation.

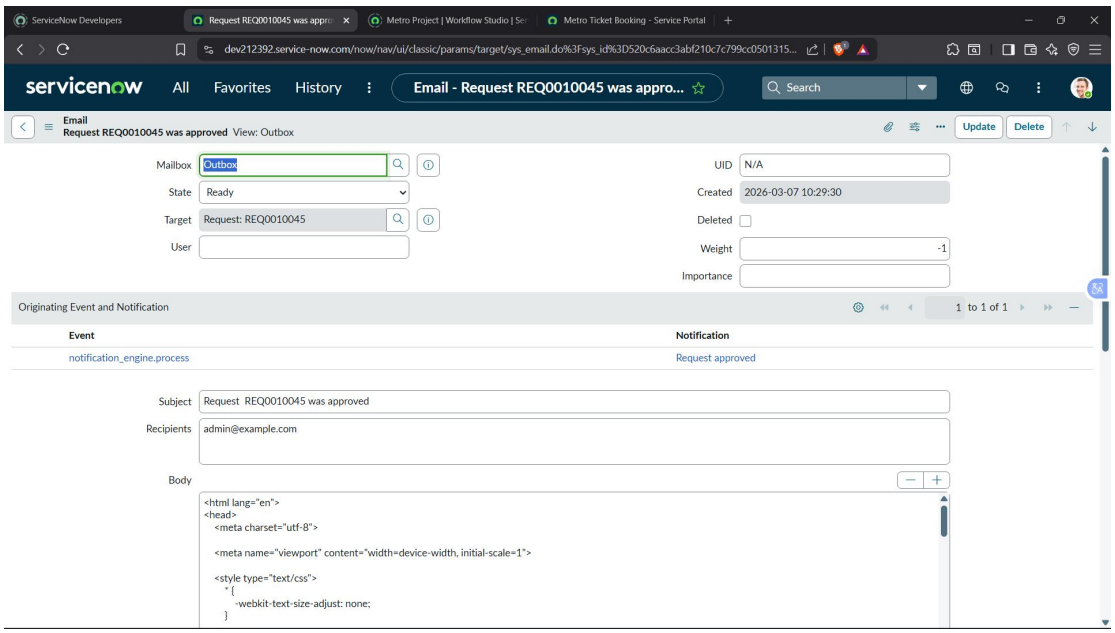
The tested workflow included:

User opens the Service Portal.
User selects Book A Metro Ticket.
User enters travel details.
Fare is calculated automatically.
User submits the request.
System creates a ticket record in the database.
QR code is generated and displayed to the user.

The system successfully completed the entire process without errors, confirming that all modules are functioning correctly.

Ticket Sent over Email - Final Confirmation





Benefits of QA Testing

QA testing helped ensure that the Metro Ticket System operates reliably and efficiently. The testing process confirmed that:

The ticket booking form works correctly
Fare calculation logic is accurate
Ticket records are stored properly in the database
QR codes are generated successfully
The entire booking workflow operates smoothly

This testing phase ensured that the system is ready for deployment and can provide a reliable ticket booking experience for users.

Data Integrity

Data integrity ensures that the information stored in the Metro Ticket Generating System remains accurate, consistent, and reliable throughout the ticket booking process. Maintaining data integrity is essential to prevent incorrect ticket information, incomplete submissions, or invalid travel details.

In this project, multiple validation and automation mechanisms were implemented to maintain the correctness of the data entered by users. These mechanisms ensure that all required fields are properly filled, correct fare values are calculated, and only valid travel details are stored in the database.

By implementing validation rules, mandatory field checks, and automated field mapping, the system guarantees that ticket records are stored correctly and consistently.

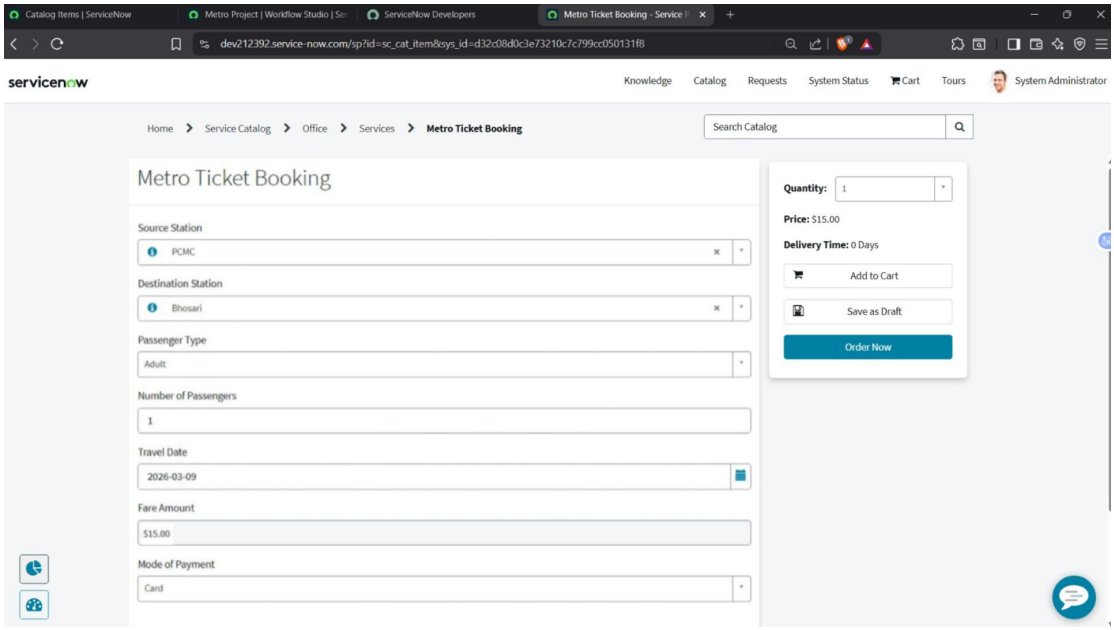
Field Validation

Field validation was implemented to ensure that users provide valid inputs when booking a metro ticket. The system checks the entered values before allowing the ticket request to be submitted.

The following validations were implemented:

Source station must be selected
Destination station must be selected
Source and destination stations cannot be empty
Travel date must be valid
Fare amount must match the calculated value

These validations prevent incorrect or incomplete ticket information from being stored in the system.



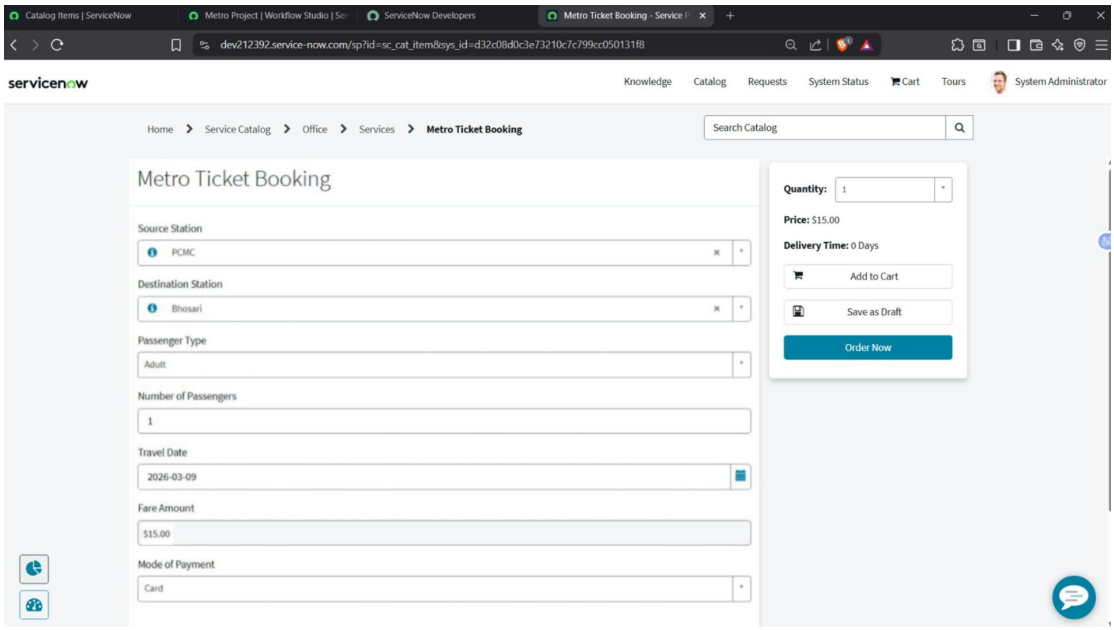
Mandatory Field Enforcement

Certain fields were configured as mandatory fields to prevent users from submitting incomplete ticket requests.

The mandatory fields include:

Source Station
Destination Station
Passenger Type
Number of Passengers
Travel Date

Both client-side validation (Catalog Client Scripts) and server-side processing (Flow Designer) ensure that these fields must be completed before the request is submitted.



Auto-Populate Logic

Automation was implemented to automatically populate certain fields in the ticket booking form. This helps reduce manual input errors and improves the user experience.

Examples of auto-populated values include:

Default journey type (Single or Return)
Automatic fare calculation based on selected stations
Automatic ticket status generation

These values are populated using Catalog Client Scripts and UI Policies, which dynamically update the form based on user selections.

Automated Field Mapping

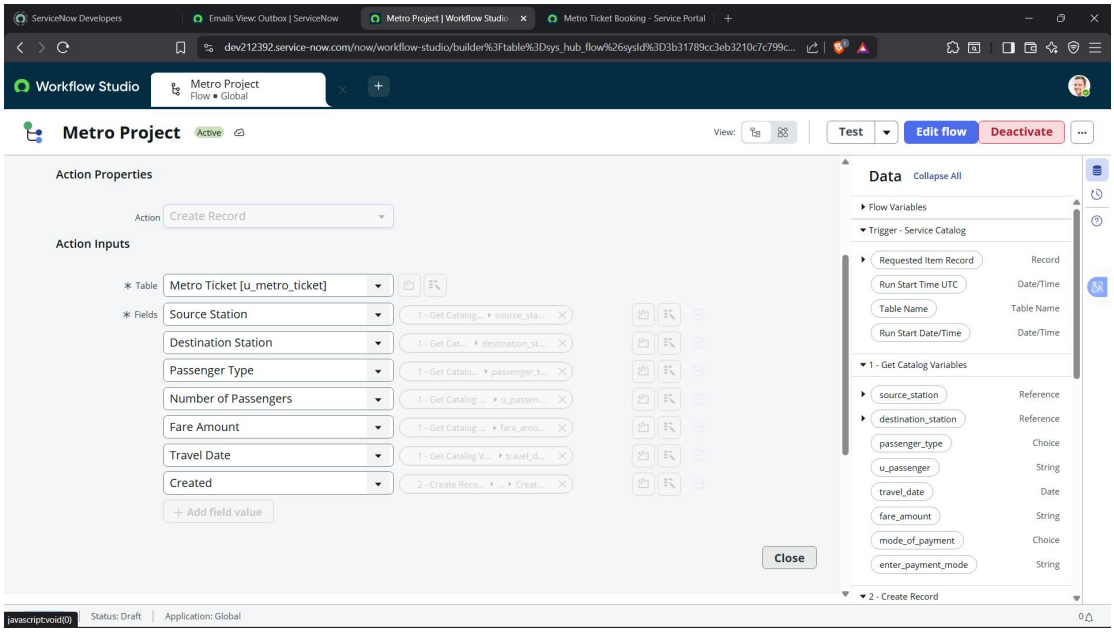
To ensure accurate data storage, the system automatically maps catalog variables to fields in the Metro Ticket database table.

This mapping is performed using the Process Automation Engine (Flow Designer).

The automation performs the following steps:

Retrieves user inputs from catalog variables.
Maps those variables to corresponding fields in the custom table.
Creates a new ticket record in the Metro Ticket database.

This ensures that all submitted ticket data is stored correctly and consistently in the database.



Benefits of Data Integrity Implementation

Implementing data integrity controls provides several benefits to the Metro Ticket System:

Prevents incomplete or incorrect ticket submissions
Ensures accurate fare calculations
Maintains consistent ticket records in the database
Improves system reliability and user trust
Supports secure and structured data management

These validation and automation mechanisms ensure that the system maintains high-quality and reliable ticket booking data.